

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

May 2028
College Park, MD

Projects

IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Engineered a C++17 order book and matching engine that replayed **50.9M** real IEX DEEP+ messages with no resync, matching 99,122 of 99,123 fills exactly and 100% of top-of-book prices
- Benchmarked B+ tree vs LSM skip-list stores for the simulation's event logs on identical 2M-record workloads; the B+ tree gave **12.8×** read throughput and **22×** faster range summaries for 27% more disk
- Published an adaptive batch-sizing policy as a negative result (+71% disk, +23% node hops, -32% append rate, no query win): cost is dominated by node lookup, not batch size
- Sustained **1.26M events/sec** end-to-end; the authoritative matching engine averages 686.7 ns/event at a 1.40 GHz base clock, 1.41× a passive mirror book (485.6 ns)
- Traced a 1.65× phantom regression to a CPU-governor artifact using perf counters and a null control book; core pinning and a fixed clock cut run-to-run variance from ~6% to 0.06%

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Writing Java policies that tune upper/lower bounds on EC2 capacity reserved for Amazon Redshift operations like patching, balancing availability against cost; initial backtests complete

Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Precomputed a tree-style search index over popular workflows from a 270K+ row PostgreSQL table (Python/SQLAlchemy, Redis cache) to minimize DB queries under access limits
- Cut LLM workflow generation time **65%** to under 15s by decomposing a Claude skill into sub-skills backed by Python search/validation tools, guided by CloudWatch + OpenSearch latency logs

UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda
- Automated fine-tuning of a Bedrock source-relevancy model on an EventBridge schedule, gated to skip redundant retraining runs, with X-Ray tracing and SNS alerts on failure

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- Built a Python pipeline for PhysTwin, a PhD-led digital twin targeting ICLR, running AnyGrasp on multi-camera images and rendering grasps as Open3D overlays for a VLM; tuned its detection prompts

TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Co-founded a 110+ member campus product studio that has shipped 5 live products, including Tortuga, a class scheduler with **20,000+ visits** in its first 30 days; built Redis caching for its Prisma ORM queries

Technical Skills

Languages: C++17, Python, Java, C, SQL, TypeScript

Systems & Performance: Linux perf (hardware counters, top-down analysis), POSIX, STL, B+ trees, LSM trees, Redis, PostgreSQL

Cloud & Tools: AWS (Lambda, Kinesis, Fargate, CDK, CloudWatch), Docker, GitHub Actions, Pytest

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)